

16662 – Robot Autonomy

Homework 3

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1. Propositional Logic

Move to pantry:

```
### Move to Pantry
# Robot in the kitchen and Robot not in the pantry
Precond = np.zeros([nrObjects, nrPredicates])
Precond[0][5] = -1 # Robot not in the pantry
Precond[0][1] = 1 # Robot in the kitchen

# Move robot from the kitchen to the pantry (remove from kitchen and add to pantry)
Effect = np.zeros([nrObjects, nrPredicates])
Effect[0][1] = -2.0 # Move robot out of the kitchen
Effect[0][5] = 2.0 # Move robot into the pantry

ActionPre.append(Precond)
ActionEff.append(Effect)
ActionDesc.append("Move to Pantry from Kitchen")
```

Move from pantry:

```
### Move from Pantry
# Robot not in the kitchen and Robot in the pantry
Precond = np.zeros([nrObjects, nrPredicates])
Precond[0][1] = -1 # Robot not in the kitchen
Precond[0][5] = 1 # Robot in the pantry

# Move robot from the pantry to the kitchen (remove from pantry and add to kitchen)
Effect = np.zeros([nrObjects, nrPredicates])
Effect[0][5] = -2.0 # Move robot out of the pantry room
Effect[0][1] = 2.0 # Move robot into the kitchen

ActionPre.append(Precond)
ActionEff.append(Effect)
ActionDesc.append("Move to Kitchen from Pantry")
```

Cut fruit in kitchen:

```
###Cut fruit in kitchen
for j in [1, 2]:
    # Robot in the kitchen, fruit in the kitchen, knife in the kitchen, fruit not chopped
    Precond = np.zeros([nrObjects, nrPredicates])
    Precond[0][1] = 1 # Robot in the kitchen
    Precond[j][1] = 1 # Fruit in the kitchen
    Precond[4][1] = 1 # Knife in the kitchen
    Precond[j][6] = -1 # Fruit not chopped

    # Fruit is chopped
    Effect = np.zeros([nrObjects, nrPredicates])
    Effect[j][6] = 2.0 # Fruit is chopped

    ActionPre.append(Precond)
    ActionEff.append(Effect)
    ActionDesc.append("Cut " + Objects[j] + " in the kitchen")
```

2. Discrete Search

A*:

```
● abhinandans-mbp:16_662_HW3 abhi$ python3 SymDiscreteSearch.py
Path Found: True
States Explored: 2005
Plan Length: 16

Plan:
Move to InKitchen from InHallway
Move to Pantry from Kitchen
Pick up Lemon from InPantry
Move to Kitchen from Pantry
Move to InHallway from InKitchen
Move to InGarden from InHallway
Pick up Strawberry from InGarden
Place Lemon at InGarden
Move to InHallway from InGarden
Move to InOffice from InHallway
Pick up Knife from InOffice
Move to InHallway from InOffice
Move to InKitchen from InHallway
Place Strawberry at InKitchen
Place Knife at InKitchen
Cut Strawberry in the kitchen
```

Dijkstra:

```
● abhinandans-mbp:16_662_HW3 abhi$ python3 SymDiscreteSearch.py
Path Found: True
States Explored: 5894
Plan Length: 16

Plan:
Move to InKitchen from InHallway
Move to Pantry from Kitchen
Pick up Lemon from InPantry
Move to Kitchen from Pantry
Move to InHallway from InKitchen
Move to InOffice from InHallway
Pick up Knife from InOffice
Move to InHallway from InOffice
Move to InGarden from InHallway
Pick up Strawberry from InGarden
Place Lemon at InGarden
Move to InHallway from InGarden
Move to InKitchen from InHallway
Place Strawberry at InKitchen
Place Knife at InKitchen
Cut Strawberry in the kitchen
```